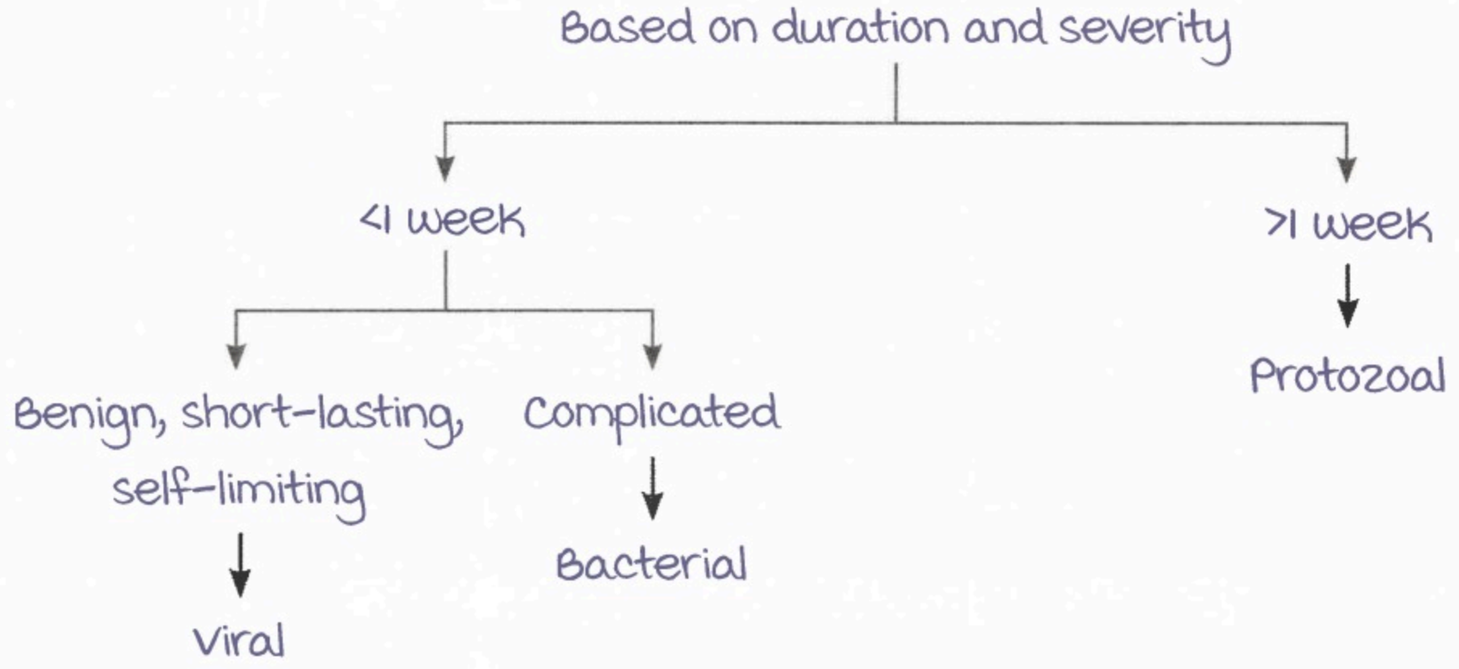
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INFECTIOUS DIARRHOEA

Introduction To Infectious Diarrhea 00:00:40

Acute diarrhea (<2 weeks) : Considered infectious unless proven otherwise.

Types of diarrhea :



Based on site :

1. Small bowel.
2. Large bowel.
3. Enterocolitis (Small bowel + Large bowel) :
   - Clostridium difficile.
   - Yersinia : Severe ileocolitis (Right lower quadrant tenderness → mimics appendicitis).

	Small bowel	Large bowel
Clinical features	Stools : • Large volume • Watery (No blood/pus/mucus/ ↑ frequency/ urgency/tenesmus)	• Fever • Small volume bloody diarrhea • Abdominal pain • Fecal leukocytes
Virus	• Norvovirus (m/c in adults) • Rotavirus (m/c in children) • Cytomegalovirus (m/c in post transplant) : Sore throat, diarrhea, leucopenia.	Cytomegalovirus (Large bowel > Small bowel)

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	Small bowel	Large bowel
Bacteria	Secretory diarrhea : • In vivo toxin production (↑cAMP → ↑water drawing → Secretory diarrhea). – ETEC (Enterotoxigenic Escherichia coli) : m/c. – Vibrio cholera. • Preformed toxins (Food poisoning) : – Staphylococcus aureus. – Bacillus cereus. – Clostridium perfringens.	Invasive organisms : • Shigella flexneri (10^2 organisms) : Causes shigellosis (m/c). • Campylobacter. • Non-typhoid salmonella. • EIEC (Enteroinvasive Escherichia coli). • Vibrio parahemolyticus.
Protozoa	• Cryptosporidium. • microsporidium. • Giardia lamblia.	Entamoeba histolytica.

Note :

Salmonella typhi : Typhoid (Small bowel invasion → Fever with chills, pain abdomen, no diarrhea).

Small Bowel Diarrhea

00:10:45

STAPHYLOCOCCUS AUREUS

Preformed toxin : Heat stable enterotoxin.

Incubation period : 1–6 hours.

Source : Custard, pork, canned meat.

Clinical features :

- Vomiting (most prominent) : D/t 5-HT3 stimulation.
- Abdominal cramps.

Note :

Organisms producing heat stable enterotoxin :

- Staphylococcus aureus.
- Bacillus cereus.
- Yersinia.
- E. coli.

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BACILLUS CEREUS

	Heat stable toxin	Heat labile toxin
Toxin	Preformed	Formed
Incubation Period	1-6 hours	6-12 hours
Clinical features	Vomiting	Diarrhea (↑ cAMP)
Source	Undercooked fried rice	meat, pudding, dried potato

VIBRIO CHOLERA

Pathogenesis :

- Non-invasive.
- Enterotoxin → Activates adenyl cyclase.
- Toxin co-regulated pilus : mediates attachment to intestinal mucosa.

Clinical features :

- Watery diarrhea : Large volume.
- Dehydration.
- Hypokalemic acidosis (Normal anion gap metabolic acidosis).

Treatment :

- Ringer lactate (Fluid of choice).
- Drug of choice :
  - Doxycycline.
  - Azithromycin (In pregnancy).

ENTEROTOXIGENIC E. COLI (ETEC)

m/c cause of community acquired diarrhea, traveller's diarrhea.

Toxin :

- Heat labile toxin (LT) → ↑ cAMP.
- Heat stable toxin of ETEC ↑ cGMP.

Note :

Vibrio Parahemolyticus :

Source : Raw fish.

Clinical features : Acute diarrhea.

Diagnosis : Kanagawa phenomenon. No role of antibiotics.

Enterohemorrhagic E.coli (EHEC) : Hemolytic uremic syndrome (Hemolytic anemia, thrombocytopenia).

Childhood HUS

90% :

- Enterohemorrhagic E.coli O157 : H7 (m/c); Shiga like toxin (Verocytotoxin).
- Shigella dysenteriae : Shiga toxin.

10% :

Streptococcus pneumonia : Neuraminidase toxin.

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Large Bowel Diarrhea

00:16:30

SHIGELLA

Pathogenesis :

Invasion by plasmid antigen (Affects all 4 layers of intestinal wall)

- Shigellosis (Severe inflammatory diarrhea).
- Reactive arthritis.

Toxin mediated

- Enterotoxin → Diarrhea
- Shiga like toxin → Cytotoxic

Treatment :

- Ciprofloxacin (DOC).
- Azithromycin.

Note :

D/d for diarrhoea with arthritis :

Inflammatory Bowel Disease (IBD) : Identified by

a. Fecal lactoferrin / calprotectin.

b. Extraintestinal manifestations:

- Erythema nodosum.
- Episcleritis.
- Type-I (Peripheral) arthritis.

NON-TYPHOID SALMONELLA

Source : Feco-oral.

Incubation : 6-72 hours.

Pathogenesis : Invasive enterotoxin.

Complication :

- Aortoarteritis (most severe).
- Endovascular infection.



Rose spots of enteric fever

----- Active space -----

Note :

Clinical features of typhoid (Salmonella typhi) :

- Enteric fever : Diarrhea in 2nd week.
- Perforation : Common complication.
- Bone marrow suppression.
- Intestinal hemorrhage.



Ileal perforation

CAMPYLOBACTER JEJUNI

Source : Feco-oral.

Incubation period : 2-4 days.

Complications :

- Risk factor for Inflammatory bowel disease.
- Trigger for GBS (Guillain Barre syndrome).
- Associated with immune proliferative small intestinal disease (IPSID)/lymphoma.

MANAGEMENT

- Indications :
  - >70 yrs.
  - Fever.
  - Abdomen pain.
  - Hypovolemia.
  - ≥6 stools/day.
- Investigation : Culture (Shigella, Salmonella, Campylobacter).
- Treatment : Azithromycin 500mg.

} Not routinely indicated in small bowel diarrhoea.

Pseudomembranous Enterocolitis 00:23:10

Etiopathogenesis :

- Organism : Clostridium difficile.
- Antibiotics (2 weeks) : Change gut flora
  - Fluoroquinolone.
  - Cephalosporin (m/c seen).
  - Clindamycin (Highest correlation).
  - Amoxicillin.
  - Ampicillin.

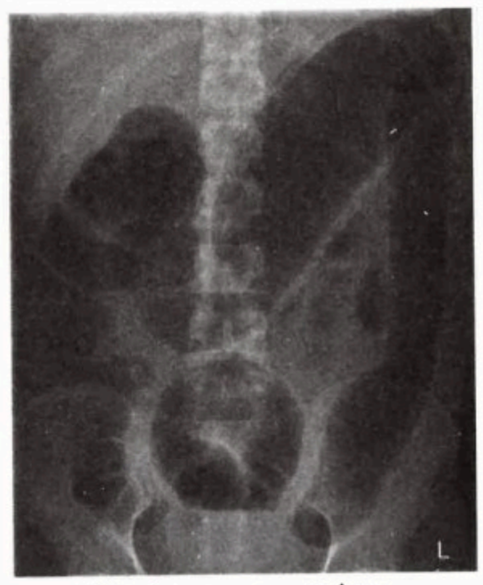
Toxins :

- Toxin A : Enterotoxin.
- Toxin B : Cytotoxic (most potent).

Incubation period : 2 days.

Clinical features :

- Fever.
- Abdominal cramps.
- Watery diarrhea with mucus +/- blood (Large bowel pattern).


Toxic megacolon

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Complication :

Toxic megacolon.

(Pseudomembranous enterocolitis > Ulcerative colitis > Infection of invasive organisms).

Diagnosis :

Glutamate dehydrogenase test (GDH) + Enzyme Immunosorbent assay (EIA) against toxin A/B.

- Both positive → Confirmed diagnosis
- Any 1 positive → PCR (Nucleic acid amplification test) → Confirmed diagnosis
- Both negative → Clostridium difficile infection ruled out.

Management :

Assess Severity

- WBC < 15000
  - S. Creatinine < 1.5
    - Non-Severe
      - Oral Fidaxomicin : DOC (↓ recurrence)
      - Oral Vancomycin
- WBC > 15000
  - S. Creatinine > 1.5
    - Ileus, Shock, megacolon
      - No → Severe
        - Oral Fidaxomicin 200mg BD
        - Oral Vancomycin
      - Yes → Fulminant
        - Oral Vancomycin
        - IV metronidazole
        - Surgical consultation